Indian Institute of Technology (Indian School of Mines) Dhanbad

Assignment-2, Max. Marks: 10

Course Code & Name: NCHS201, Chemical Plant Design and Economics

Instructor: Dr. Ejaz Ahmad

Make suitable assumptions wherever necessary

Q1. Create an Excel sheet using the Solver/Seek function for the estimation of capital investment and for the estimation of product cost.

Table 6-17 Estimation of capital investment cost (showing individual components)

The percentages indicated in the following summary of the various costs constituting the capital investment are approximations applicable to ordinary chemical processing plants. It should be realized that the values given vary depending on many factors, such as plant location, type of process, and complexity of instrumentation.

I. **Direct costs** = material and labor involved in actual installation of complete facility (65–85% of fixed-capital investment)
   - A. Equipment + installation + instrumentation + piping + electrical + insulation + painting (50–60% of fixed-capital investment)
     1. Purchased equipment (15–40% of fixed-capital investment)
     2. Installation, including insulation and painting (25–55% of purchased-equipment cost)
     3. Instrumentation and controls, installed (8–50% of purchased-equipment cost)
     4. Piping, installed (10–80% of purchased-equipment cost)
     5. Electrical, installed (10–40% of purchased-equipment cost)
   - B. Buildings, process, and auxiliary (10–70% of purchased-equipment cost)
   - C. Service facilities and yard improvements (40–100% of purchased-equipment cost)
   - D. Land (1–2% of fixed-capital investment or 4–8% of purchased-equipment cost)

II. **Indirect costs** = expenses which are not directly involved with material and labor of actual installation of complete facility (15–35% of fixed-capital investment)
   - A. Engineering and supervision (5–30% of direct costs)
   - B. Legal expenses (1–3% of fixed-capital investment)
   - C. Construction expense and contractor's fee (10–20% of fixed-capital investment)
   - D. Contingency (5–15% of fixed-capital investment)

III. **Fixed-capital investment** = direct costs + indirect costs

IV. **Working capital** (10–20% of total capital investment)

V. **Total capital investment** = fixed-capital investment + working capital

Table 6-18 Estimation of total product cost (showing individual components)

The percentages indicated in the following summary of the various costs involved in the complete operation of manufacturing plants are approximations applicable to ordinary chemical processing plants. It should be realized that the values given vary depending on many factors, such as plant location, type of process, and company policies.

I. **Manufacturing cost** = direct production costs + fixed charges + plant overhead costs

A. Direct production costs (about 66% of total product cost)

1. Raw materials (10–80% of total product cost)
2. Operating labor (10–20% of total product cost)
3. Direct supervisory and clerical labor (10–20% of operating labor)
4. Utilities (10–20% of total product cost)
5. Maintenance and repairs (2–10% of fixed-capital investment)
6. Operating supplies (10–20% of maintenance and repair costs, or 0.5–1% of fixed-capital investment)
7. Laboratory charges (10–20% of operating labor)
8. Patents and royalties (0–6% of total product cost)

B. Fixed charges (10–20% of total product cost)

1. Depreciation (depends on method of calculation—see Chap. 7)
2. Local taxes (1–4% of fixed-capital investment)
3. Insurance (0.4–1% of fixed-capital investment)
4. Rent (8–12% of value of rented land and buildings)
5. Financing (interest) (0–10% of total capital investment)

C. Plant overhead costs (50–70% of cost for operating labor, supervision, and maintenance; or 5–15% of total product cost) include costs for the following: general plant upkeep and overhead, payroll overhead, packaging, medical services, safety and protection, restaurants, recreation, salvage, laboratories, and storage facilities

II. **General expenses** = administrative costs + distribution and selling costs + research and development costs (15–25% of the total product cost)

A. Administrative costs (about 20% of costs of operating labor, supervision, and maintenance; or 2–5% of total product cost) include costs for executive salaries, clerical wages, computer support, legal fees, office supplies, and communications

B. Distribution and marketing costs (2–20% of total product cost) include costs for sales offices, salespeople, shipping, and advertising

C. Research and development costs (2–5% of every sales dollar, or about 5% of total product cost)

III. **Total product cost** = manufacturing cost + general expenses

IV. **Gross earnings cost** (gross earnings = total income − total product cost; amount of gross earnings cost depends on amount of gross earnings for entire company and income tax regulations; a general range for gross earnings cost is 15–40% of gross earnings)